Case Study 3

Predicting Titanic Passenger Survival using Logistic Regression

Title of case study	Logistic Regression: Titanic Survival Prediction
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Section	CSE-3
Year	3rd Year, 5th Semester
Subject name	Fundamentals of Data Science
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1. Problem Statement

The sinking of the RMS Titanic in 1912 killed most of the people aboard, and survival was far from random: passengers' sex, class and age appear to have influenced who reached a lifeboat. The objective of this case study is to build a **binary classifier** that predicts whether a passenger survived (*Survived* = 1) or not (0) from the recorded passenger attributes, to evaluate how reliable that prediction is, and to interpret which attributes were most strongly associated with survival.

The dataset contains 891 passengers and 12 columns: *PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked*. Logistic regression is the chosen method because the outcome is binary and the model's coefficients can be read directly as effects on the odds of survival, which makes the results explainable.

2. Methodology

2.1 Data exploration

Exploratory data analysis covered the feature types, summary statistics, missing values, the distribution of the target, and the relationship between survival and each feature using bar charts, histograms, box plots and a correlation matrix.

2.2 Data preprocessing

- **Missing values.** *Age* (19.9% missing) was imputed with the median age of the passenger's class and sex group. *Embarked* (2 missing) was filled with the most frequent port (Southampton). *Cabin* (77.1% missing) was dropped as too sparse to impute reliably.
- **Dropped columns.** *PassengerId, Name* and *Ticket* are identifiers/free text with no direct predictive meaning for a linear model.
- **Encoding.** *Sex* was binary-encoded (female = 1). *Embarked* was one-hot encoded with Cherbourg as the baseline category.

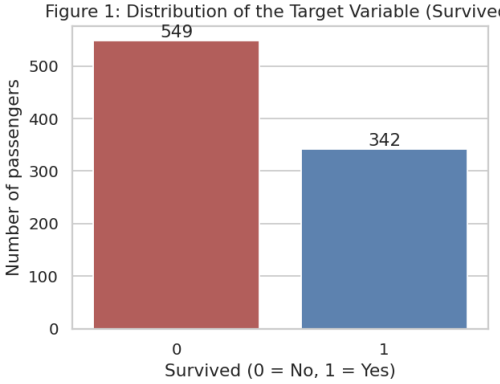
2.3 Model building

The data was split 80/20 into training (712 passengers) and test (179 passengers) sets, stratified on the target so both keep a survival rate of about 38%. A scikit-learn pipeline standardises the features (using training-set statistics only, to avoid leakage) and fits a logistic regression with default L2 regularisation. Performance was evaluated on the unseen test set and checked with 5-fold stratified cross-validation.

3. Statistical Findings (Exploratory Analysis)

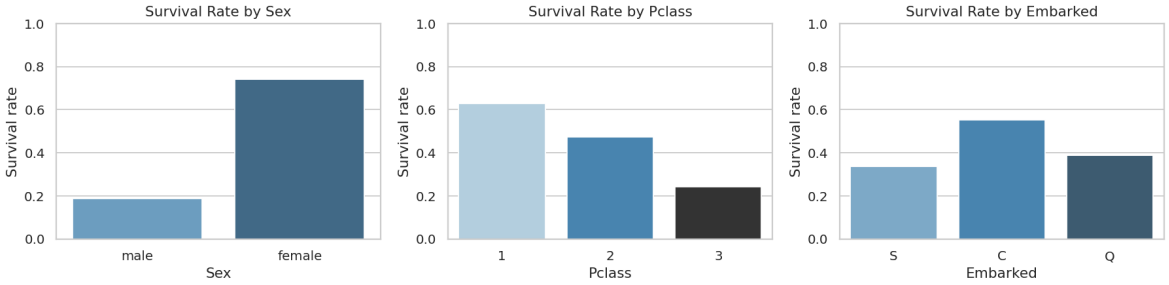
Of the 891 passengers, 342 (38.4%) survived and 549 (61.6%) did not, so the classes are moderately imbalanced. Precision, recall, F1 and ROC-AUC are therefore reported in addition to accuracy.

Figure 1: Distribution of the Target Variable (Survived)



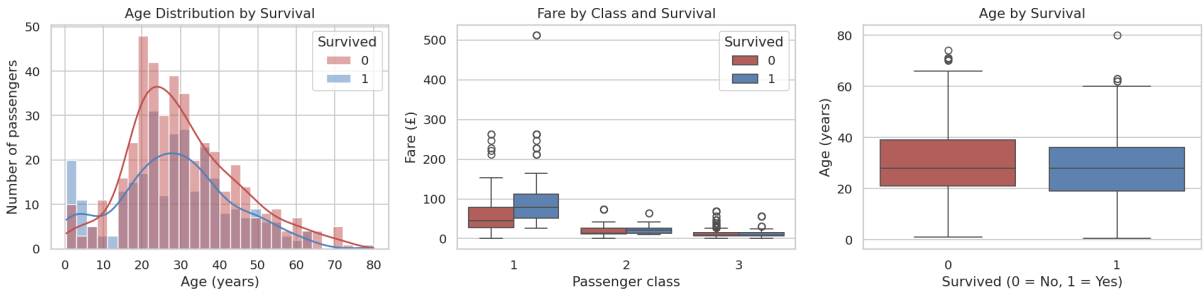
Interpretation: Fewer passengers survived than died. A model that always predicted "did not survive" would already be right about 61.6% of the time, so any useful model must clearly beat this baseline.

Figure 2: Survival Rate by Sex, Passenger Class and Port of Embarkation



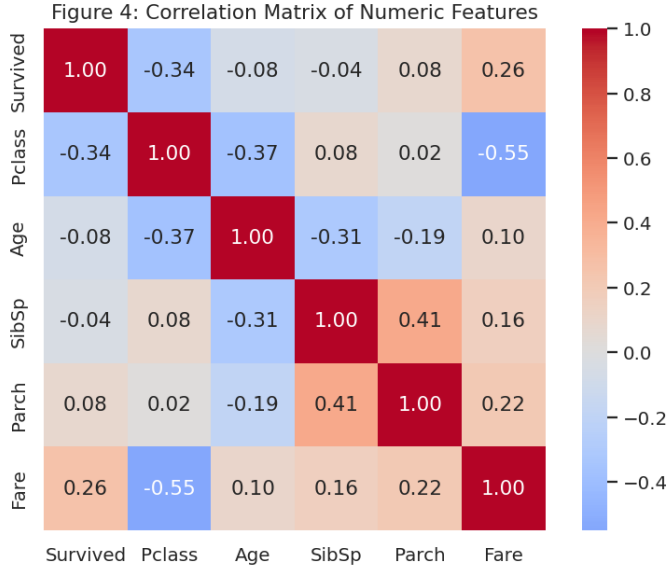
Interpretation: Sex is the dominant factor: 74.2% of women survived versus 18.9% of men. Survival also falls steadily with class (63.0% in 1st, 47.3% in 2nd, 24.2% in 3rd). Cherbourg passengers survived more often (55.4%) than those from Queenstown (39.0%) or Southampton (33.7%); this is probably partly a class effect, because the passenger mix differed by port.

Figure 3: Age and Fare Distributions



Interpretation: The age distributions of survivors and non-survivors overlap heavily and have similar medians, but there is a noticeable excess of survivors among young children, consistent with "women and children first". Fares are strongly tied to class: 1st-class fares are much higher and highly skewed (outliers above £500), while within 1st class survivors tended to have paid somewhat more.

Figure 4: Correlation Matrix of Numeric Features



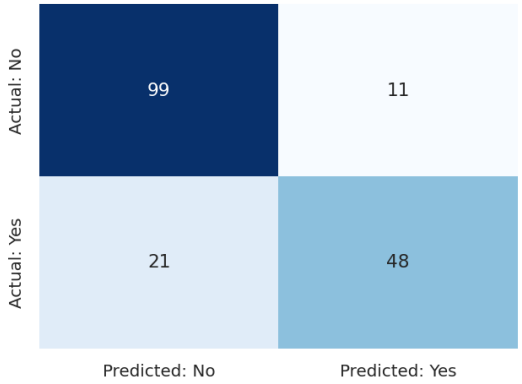
Interpretation: *Pclass* has the strongest linear association with survival (r = -0.34): higher-numbered (lower) classes survived less. *Fare* is positively associated (r = 0.26), but since fare and class carry overlapping information, much of this is a class effect. Age (r = -0.08), SibSp (r = -0.04) and Parch (r = 0.08) show only weak linear relationships. Age's effect is concentrated among children, which a single correlation coefficient understates.

4. Model Evaluation

The model was evaluated on the 179 held-out test passengers.

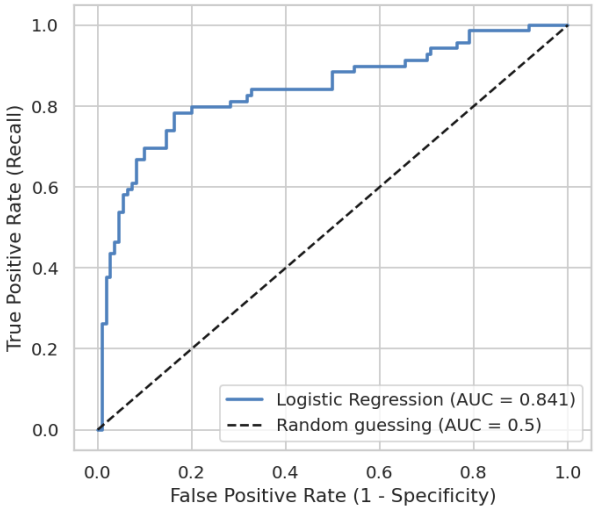
Metric	Value	Meaning
Accuracy	0.821	About 82% of test passengers classified correctly (baseline of always predicting "died": 61.5%).
Precision	0.814	When the model predicts survival, it is right about 81% of the time.
Recall	0.696	The model finds about 70% of the actual survivors.
F1-score	0.750	Balance of precision and recall for the survivor class.
ROC-AUC	0.841	A random survivor is ranked above a random non-survivor 84% of the time.

Figure 5: Confusion Matrix (Test Set)



Interpretation: Of 110 passengers who died, 99 were correctly identified (11 false alarms). Of 69 survivors, 48 were identified but 21 were missed. The model is more reliable at recognising non-survivors (recall 0.90) than survivors (recall 0.70), i.e. it is somewhat conservative about predicting survival.

Figure 6: ROC Curve (Test Set)



Interpretation: The ROC curve bends well above the diagonal of random guessing, giving an AUC of 0.841. The model separates survivors from non-survivors well across decision thresholds, not only at the default 0.5 cut-off.

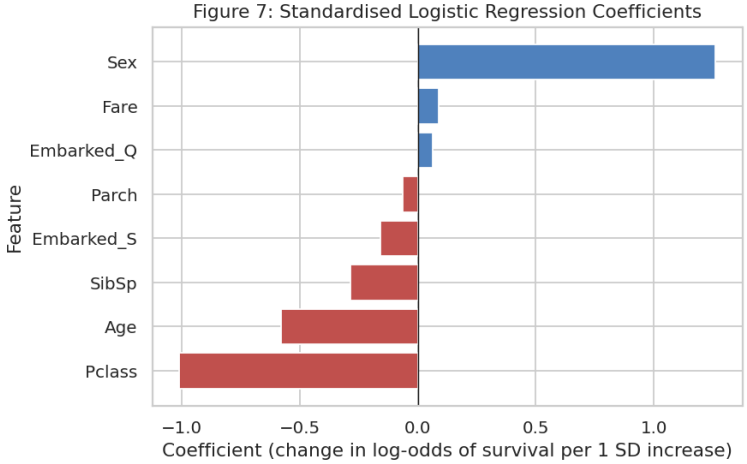
Generalisation. Training accuracy (80.8%) is close to test accuracy (82.1%), so there is no sign of overfitting. Five-fold cross-validation gave a mean accuracy of 79.9% (±1.3%) and a mean ROC-AUC of 0.854 (±0.022). The single test split is therefore slightly optimistic, and about 80% accuracy is the realistic expectation on new data.

5. Interpretation of the Coefficients

Logistic regression models the log-odds of survival. Exponentiating a coefficient gives an odds ratio (OR): the multiplicative change in the odds of survival for a one-unit increase in the feature, holding the others fixed. Significance was assessed with Wald z-tests on an essentially unregularised fit of the training data.

Feature	Odds ratio (95% CI)	p-value	Interpretation
Sex (female = 1)	14.5 (9.3 – 22.5)	< 0.001	Women have about 14 times the odds of survival of men of the same class, age and family size.
Pclass	0.29 (0.21 – 0.40)	< 0.001	Each step down in class cuts the odds of survival by about 70%.
Age	0.96 (0.94 – 0.97)	< 0.001	Each extra year lowers the odds by about 4%; 10 years older is about 35% lower odds.
SibSp	0.76 (0.60 – 0.96)	0.023	Each additional sibling/spouse aboard lowers the odds by about 24%.
Parch	0.92 (0.72 – 1.19)	0.53	Not significant.
Fare	1.00 (1.00 – 1.01)	0.55	Not significant once class is accounted for.
Embarked_Q (vs C)	1.28 (0.53 – 3.08)	0.58	Not significant.
Embarked_S (vs C)	0.70 (0.42 – 1.18)	0.18	Not significant.

Figure 7: Standardised Logistic Regression Coefficients



Interpretation: On the standardised scale, *Sex* (+1.26) is the most influential predictor, followed by *Pclass* (-1.01), *Age* (-0.58) and *SibSp* (-0.29). Fare, Parch and the port of embarkation contribute little. Sex, class, age and family size are therefore the features that matter for predicting survival.

Caveats. Odds ratios describe odds rather than probabilities (an odds ratio of 14 does not mean women were 14 times as likely to survive), and the results are associations in historical data, not causal effects. *Pclass* is treated as numeric (equal effect per class step) and *Age* as linear, although the exploration suggested a stronger effect for young children.

6. Conclusion

A logistic regression model predicted Titanic survival with 82.1% test accuracy, an F1-score of 0.75 and a ROC-AUC of 0.84, clearly outperforming the 61.5% majority-class baseline. Cross-validation (about 80% accuracy, AUC 0.85) confirms the result is stable and not caused by overfitting.

The model shows that being female, travelling in a higher class, being younger and having fewer siblings/spouses aboard were each associated with higher odds of survival, while fare, parents/children aboard and port of embarkation added no significant information once those factors were accounted for. The main weakness is recall: about 30% of survivors were missed. Possible improvements are a child indicator or non-linear age terms, features engineered from passenger titles and family size, and comparison with tree-based models.

The trained model was saved and deployed as a Streamlit app (*app.py*) in which a user enters passenger details and receives a predicted survival probability. The code is available in the accompanying Jupyter Notebook and GitHub repository.